

COM-480 Milestone 2 - Stockswise

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1 Project Goal

The goal of this project is to provide a data-driven experience for students and individuals who want to start learning finance concepts and begin thinking about investing their money without having a financial background. The platform focuses on intuitive and educational visualizations to break down complex and often unintuitive financial ideas. It is designed to be accessible to complete beginners while still offering depth and value for more advanced users.

2 Outline

The website is structured in two parts. The first part is dedicated to instruction and theoretical knowledge for finance beginners who want to dive into the investment world. The course is divided into modules, and each concept is supported by its own tailored visualizations, always based on our initial dataset. Each module has its own quiz, enabling users to progressively test their knowledge.

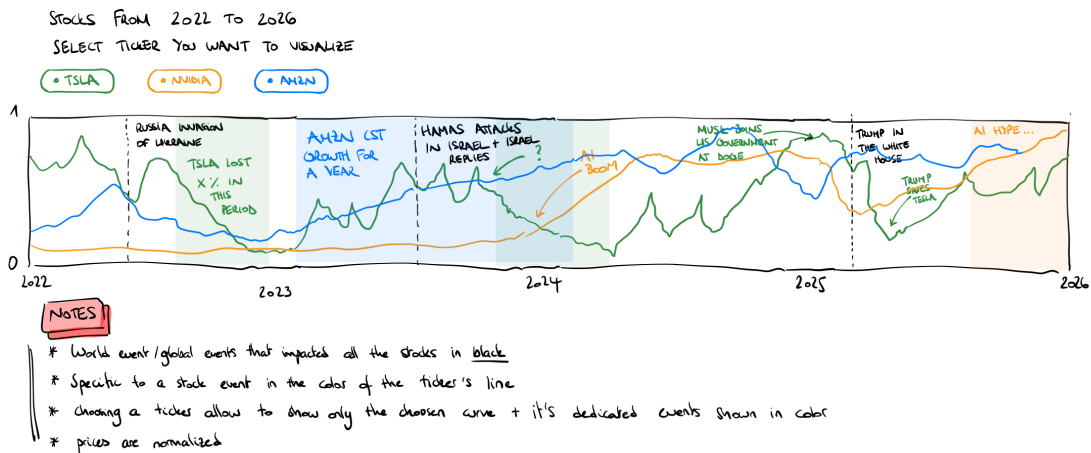


Figure 1: A visualization illustrating Module 1, showing how prices are determined and how they fluctuate over time. The full time range will not be displayed at once; instead, users can navigate through it using a slider.

Module 1 introduces stocks, what they are, how prices are set through supply and demand, and key vocabulary such as tickers, market cap, and adjusted close. Module 2 covers ETFs, explaining how they provide instant diversification compared to individual stocks, with a side-by-side performance comparison. Module 3 addresses risk, teaching users about volatility, max drawdown, and the sharpe ratio through interactive charts. Finally, Module 4 links to the simulation lab, where users can put their newly acquired knowledge into practice.

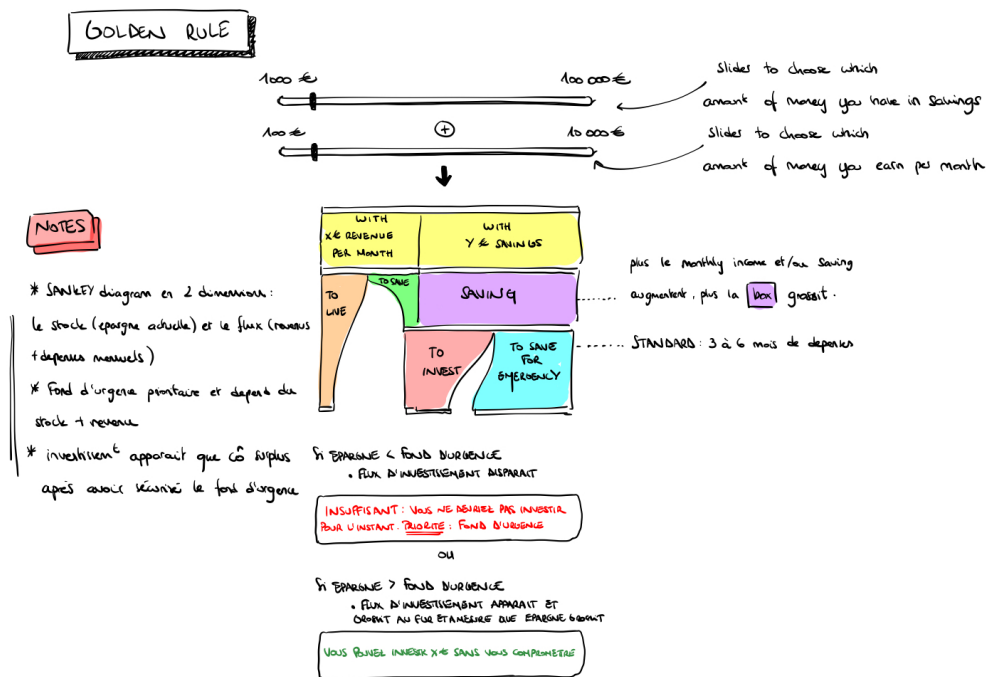


Figure 2: This figure presents an alternative view of the fundamental principle to follow when starting to invest. Sliders enable users to dynamically adjust the plots based on different budget levels.

The second part is an investment simulator. The user picks a stock or ETF, a starting date, and an amount, then sees how that investment would have performed over time through an interactive chart. Key metrics: final value, total return, CAGR, and maximum drawdown, are displayed alongside, turning abstract market data into a tangible, personal experience.

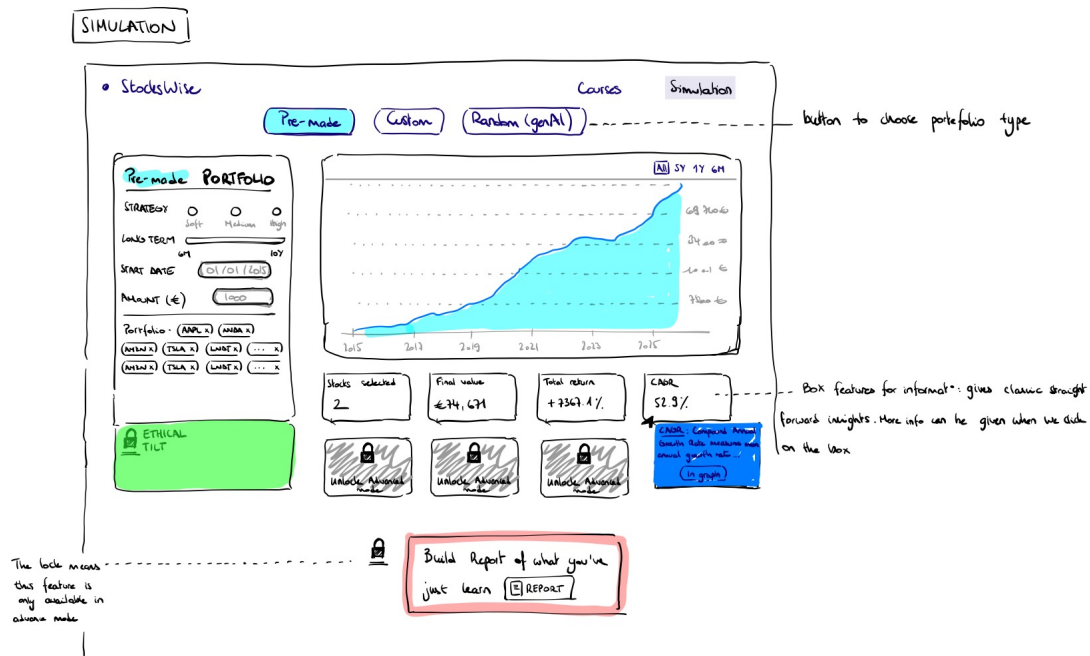


Figure 3: Example of a tailored visualization supporting a course concept.

For both parts, a toggle in the navigation bar allows switching between a beginner version and a more advanced version. (A prototype has been developed, but most advanced features are part of *Additional Ideas*, where the initial content and visualizations are adapted in terms of complexity, level of detail, and depth of analysis.) The beginner mode focuses on simplified explanations and guided interactions, while the advanced mode introduces more technical concepts, additional metrics, and richer data representations, enabling users to explore the same material from a more analytical perspective.

3 Technologies

The website structure, content, and graphics were initially sketched on an iPad. The overall prototype was then uploaded and tested online using *Claude Code*. No external frameworks were used to build the website, it relies solely on native HTML, CSS, and JavaScript for interaction. Graphics and interactive elements were developed using the JavaScript library *D3.js*.

To handle the full NASDAQ dataset (8,000 CSV files, 2.6 GB total), we opted for a separated data architecture: the files are hosted on a dedicated Git branch (data) and fetched individually on demand via the GitHub API (raw.githubusercontent.com). This approach is the most efficient as it keeps the main project branch lightweight while enabling fast loading, only the sets selected by the user are downloaded.

4 Additional Ideas

The core objective of the website is to remain as user-friendly as possible, offering an interesting user experience for beginner and more advanced people who want to understand difficult finance concept. Future development will prioritize the addition of new visualizations designed to present these concepts in a clear and intuitive manner.

These features will be integrated into an “Advanced mode,” which will include more technical tools (such as candlestick charts, indicators, and mathematical concepts), portfolio construction topics (including diversification, correlation, sector allocation, and the efficient frontier), other risk metrics, and dedicated content on the history and behavioral dynamics of the NASDAQ market.

This will bring more depth for users: Complex and unintuitive financial concepts will be unpack and illustrated through innovative visualizations, guiding users step by step to ensure a smooth and coherent learning experience.

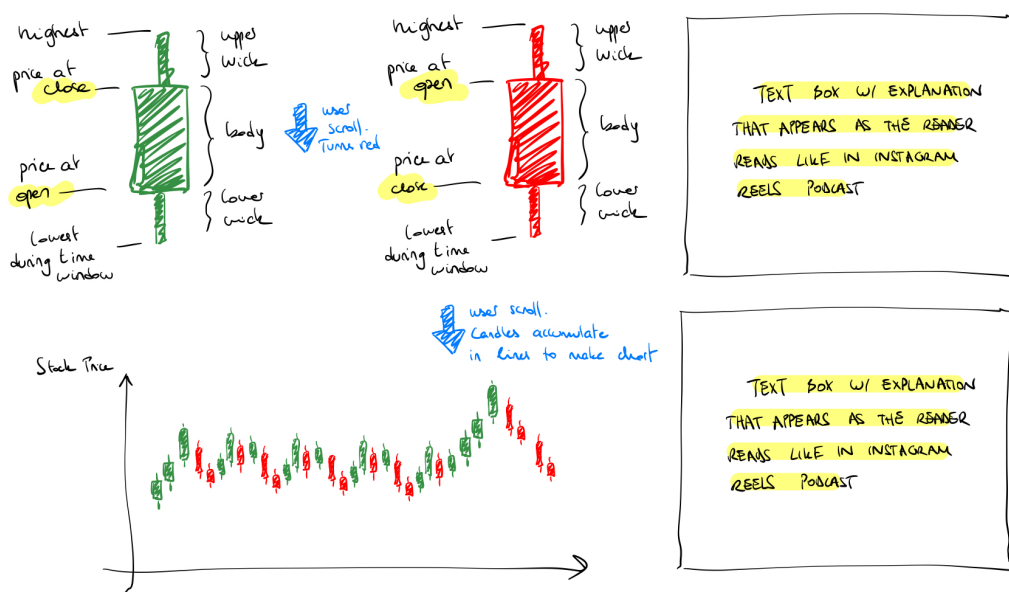


Figure 4: A visualization for the advanced mode in Module 1 that explains a candlestick as a unit and demonstrates how to interpret a standard candlestick chart. Additional indicators, such as the simple moving average (SMA), will be overlaid on the chart.